

Rotation Matrices Part 2 and intro to HT Matrices

EGR 445/545

GVSU

Plan for today

- review:
 - LA 1 debrief
 - rotation signs and conventions
 - unit vectors as columns
 - dot products
- multiple rotations
- intro to HT matrices
- project 1 kick-off
 - two-link inverse kinematics
- LA 4 and 5

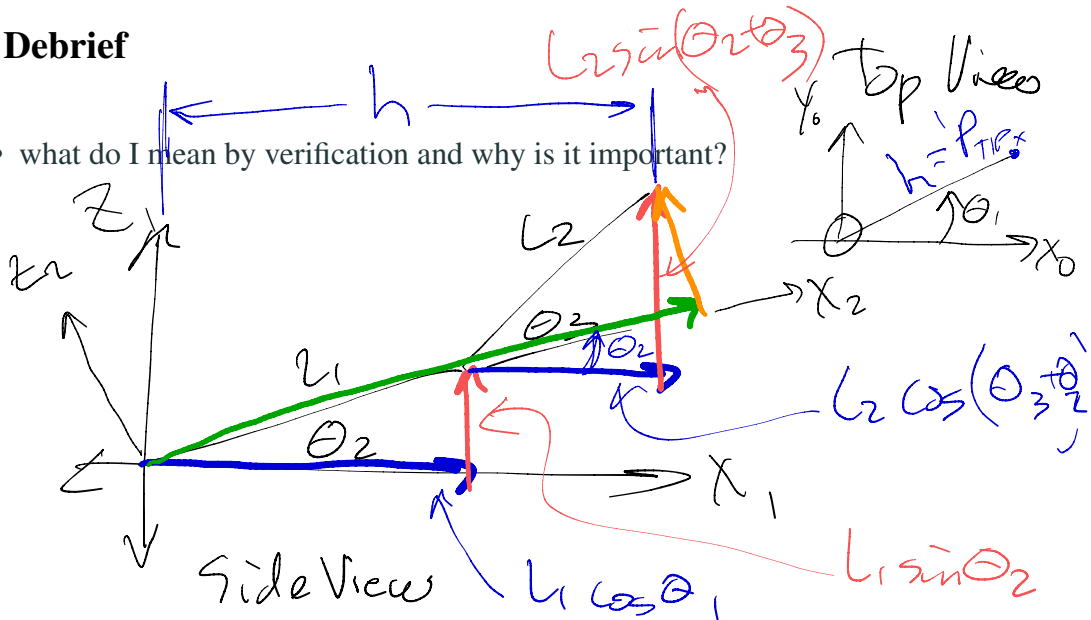
Announcements

- HW 1 is up (github)
 - dues Wed. 5/27
- Quiz 1 next Wed., 5/27  
 - what is a rotation matrix?
 - where do rotation matrices come from?
 - how do I find an expression for one based on a sketch?
 - what does it mean to say the unit vectors are the columns?
- Chapter 2 Learning Outcomes posted 

Review

LA 1 Debrief

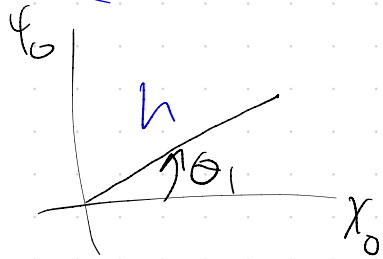
- what do I mean by verification and why is it important?



$$h = L_1 \cos \theta_2 + L_2 \cos(\theta_2 + \theta_3)$$

$${}^1P_{TIPz} = L_1 \sin \theta_2 + L_2 \sin(\theta_2 + \theta_3)$$

$${}^0P_{TIP} = \begin{bmatrix} h \cos \theta_1 \\ h \sin \theta_1 \\ {}^1P_{TIPz} \end{bmatrix}$$



Conceptual Review

- why do we need to learn rotation matrices?
- why do we also need HT matrices?

rotation + translation
in one matrix

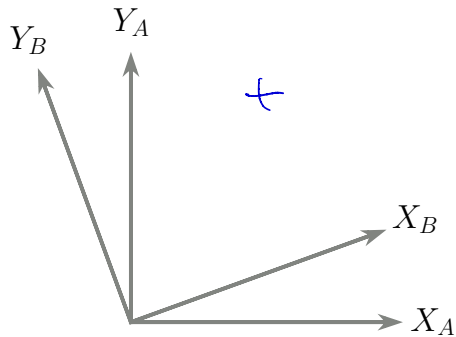
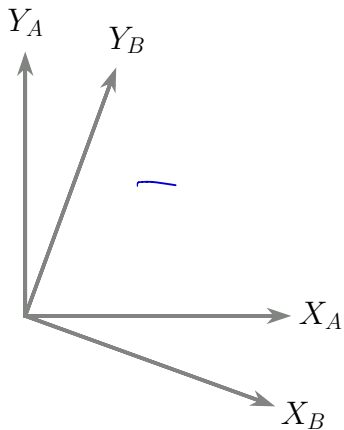
→ Go back and forth between
joint angles and tip
position and orientation
of final frame

Steps and Sign Conventions

- what are the steps and sign conventions in finding ${}^A_B\mathbf{R}$?

→ draw a sketch
→ draw a right handed
coordinate system
→ rotate frame B away from
A in a right hand sense
→ draw unit vectors in frame B
and find their components in frame A

Which of these is a positive rotation?



Dot Products

- what is the relationship between unit vectors and rotation matrices?

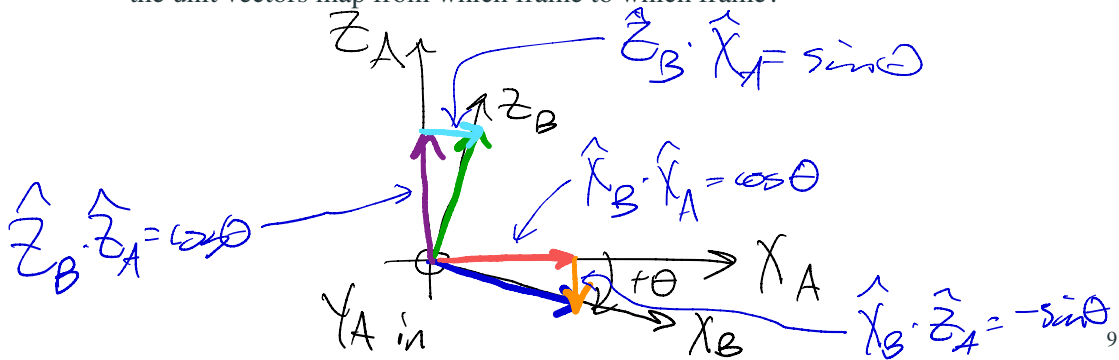
$${}^A_R_B = \begin{bmatrix} {}^A\hat{x}_B & {}^A\hat{y}_B & {}^A\hat{z}_B \end{bmatrix}$$

Frame B unit vectors expressed
in frame A components

Finding ${}^A_B\mathbf{R} = \mathbf{R}_y(\theta)$

- what sketch do we need to draw?
 - what does a positive $\mathbf{R}_y$ rotation look like?
- which unit vectors do we need to draw?
 - the unit vectors map from which frame to which frame?

$${}^A_B\mathbf{R}_y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$



Multiple Rotations: the Notation is Your Friend

- step 1: first, we rotate frame 1 from frame 0 by θ_1
- step 2: then, we rotate frame 2 from frame 1 by θ_2
- if we know 2P , how do we find 0P ?

For scalars: $2 \times 3 = 3 \times 2$ commutative

For Matrices $[A] \cdot [B] \neq [B] \cdot [A]$

Finding $\mathbf{R}$ with multiple rotations

- step 1: first, we rotate frame 1 from frame 0 by θ_1
- step 2: then, we rotate frame 2 from frame 1 by θ_2
- how do we find ${}^0_2\mathbf{R}$?

$$\cancel{{}^0_1\mathbf{R}} \cdot {}^1_2\mathbf{R} = ? \quad {}^0_2\mathbf{R}$$

$$\cancel{{}^1_2\mathbf{R}} \cdot \cancel{{}^0_1\mathbf{R}} = ? \quad ?!?$$

Multiple Rotations and Order of Rotations

Let the notation guide you

Key question:

- If we were going to rotate first from A to B and then from B to C, how would we find ${}^A_C\mathbf{R}$?
 - can our notation conventions help guide us?
 - what does the math behind ${}^A_C\mathbf{R}$ really mean?

Multiple Rotations example

1. 1st. $R_x(90)$ followed by $R_y(90)$

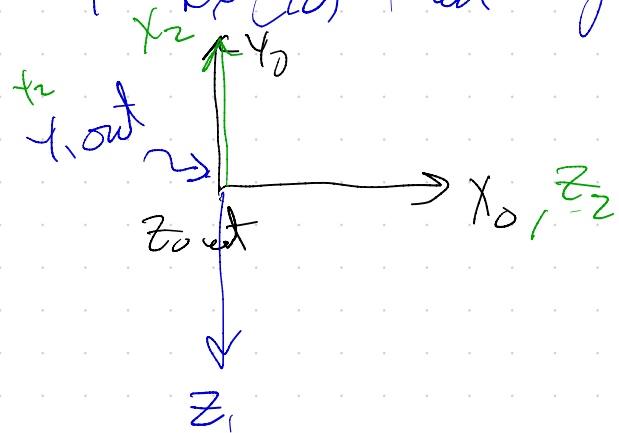
2. Compared to $R_y(90)$ followed by $R_x(90)$

→ Visualize

→ sketch

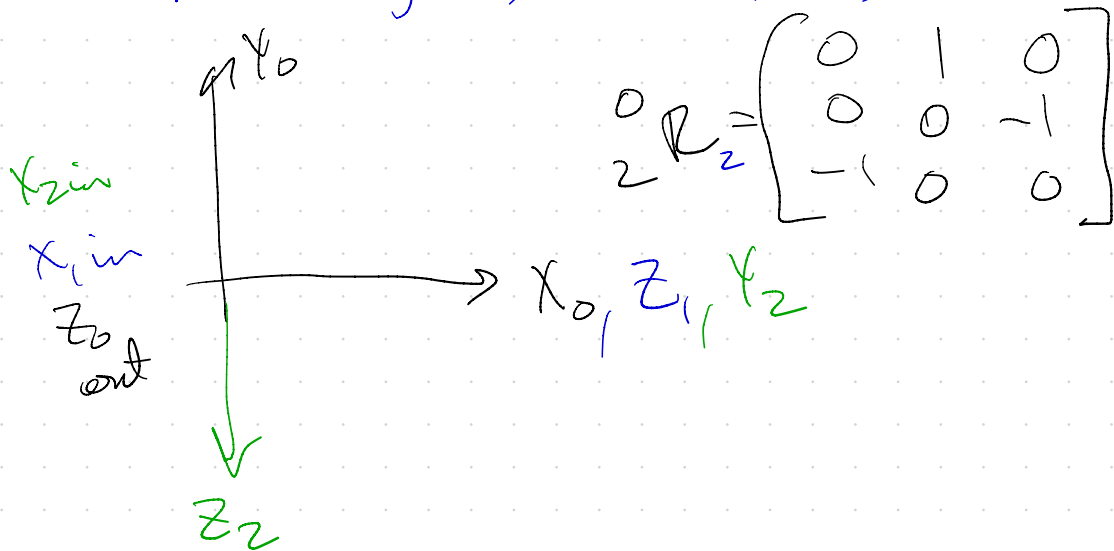
→ code

1. $R_x(90)$ then $R_y(90)$



$${}^0R_2 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Order 2 $R_y(90)$ then $R_x(90)$



Remaining Chapter 2 Topics

- how do we handle rotation and translation together?
- what is an HT matrix and how do they work?
- how do we find the inverse of an HT matrix?
- do rotation matrices really have 9 unknowns?
- constraints and Euler angles
- what are free vectors and how to we deal with them?
- computational considerations

HT

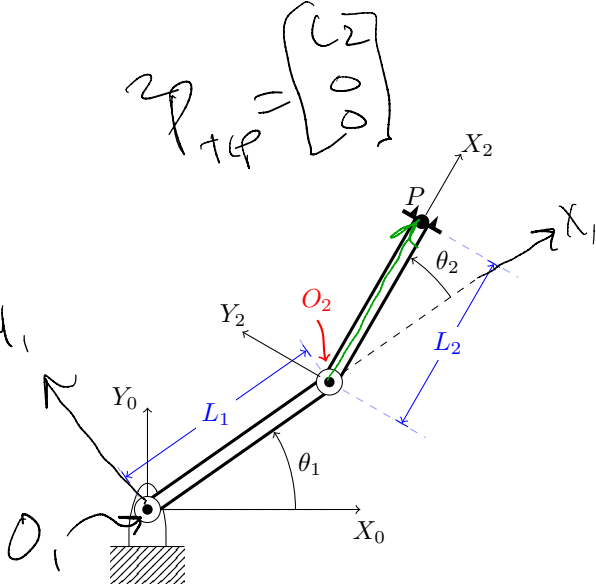
Euler

Intro to HT Matrices

Intro to HT Matrices

- How do we find ${}^1P_{tip}$?
- what is wrong with this?:

$${}^1P_{tip} = {}^1_2\mathbf{R} \cdot {}^2P_{tip} + \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix}$$

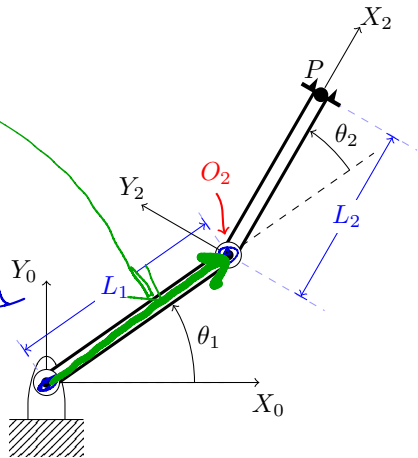


The point of an HT matrix

- how do we put this in one matrix operation?

$${}^1P_{tip} = {}^1P_{O2} + {}^1_2\mathbf{R} \cdot {}^2P_{tip}$$

a vector from O_1 to O_2
expressed in frame 1
components



The Form of an HT Matrix

- how do we add a translation and do a rotation in one step?

$${}^0_1\mathbf{T} = \begin{bmatrix} {}^0_1R & {}^0P_{orig} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Eqn 2.19

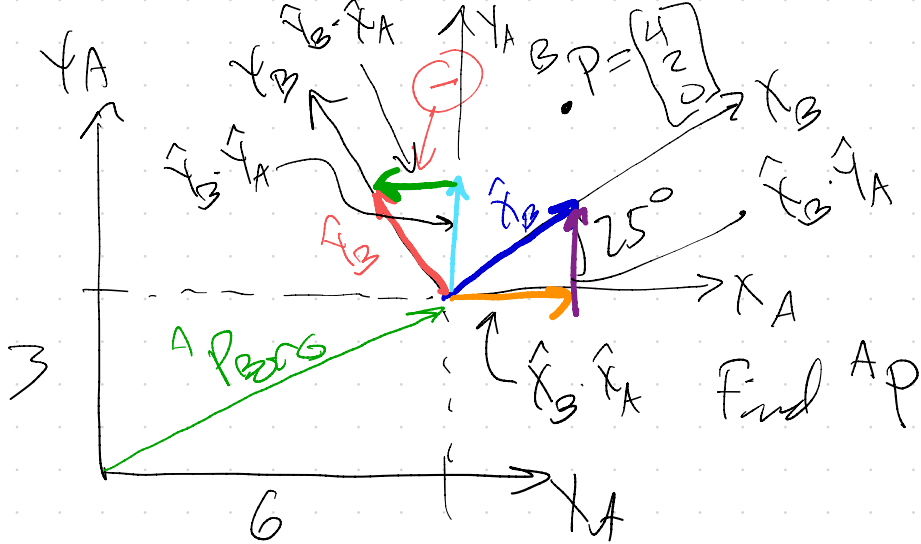
$$\begin{bmatrix} A \\ P \\ 1 \end{bmatrix} = \begin{bmatrix} A & P & \text{BORG} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ P \\ 1 \end{bmatrix}$$

Annotations:
 - Blue arrows point to the top-left and top-right blocks of the transformation matrix, both labeled 3×3 .
 - A blue arrow points to the bottom-right element of the transformation matrix, labeled 1 .
 - A blue arrow points to the bottom-left elements of the transformation matrix, labeled 0 .

$$\textcircled{A} P = \textcircled{A} P^B + \textcircled{A} P_{\text{BORG}}$$

Annotations:
 - A red circle is around the A in $\textcircled{A} P$.
 - A red circle is around the A in $\textcircled{A} P^B$.
 - A red circle is around the A in $\textcircled{A} P_{\text{BORG}}$.
 - A blue arrow points to the P^B term, labeled 1×3 .
 - A red arrow points to the P_{BORG} term, labeled 1×1 .

Addition takes place in frame A



$$AP_2 = \begin{bmatrix} \cos 25^\circ & \sin 25^\circ & 0 \\ \sin 25^\circ & \cos 25^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

AP_2

$$AP_2 = \begin{bmatrix} \cos 25^\circ & \sin 25^\circ & 0 & 6 \\ \sin 25^\circ & \cos 25^\circ & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

Project 1 Kick-Off

- projects folder under student github folder

Two-Link Inverse Kinematics

- under class folders

Active Learning

- learning activities 4 and 5